Mechanical Engineer (Physical Infrastructure)

ABOUT THE ROLE:
We are seeking exceptional Mechanical Engineers to join our growing team in Memphis and help design, build, and optimize the mechanical infrastructure for xAI's high-density AI supercomputing facilities. This is a broad Mechanical Engineer requisition covering three critical focus areas:

Thermal Mechanical Engineering (liquid cooling, heat exchangers, CFD analysis)
Systems Component Mechanical Engineering (enclosures, manifolds, structural assemblies, fluid handling)
HVAC Design Mechanical Engineering (facility air systems, CRAC/CRAH, airflow management)
You will work on one or more of these areas depending on your expertise and business needs, collaborating across thermal, mechanical, electrical, power, and construction teams to push the boundaries of performance, efficiency, and reliability at massive scale. This is a full-time, onsite role based in the Memphis region.

RESPONSIBILITIES:
Design and optimize mechanical systems for high-density AI data centers, including liquid cooling loops, cold plates, manifolds, heat exchangers, server enclosures, structural supports, and facility HVAC systems.
Perform detailed engineering analysis — thermal, fluid dynamics, structural, airflow, and heat transfer — using CFD, FEA, and other simulation tools.
Develop specifications, 3D models, drawings, BOMs, and test plans; support prototyping, qualification, and iterative improvements.
Integrate mechanical systems with power generation, electrical infrastructure, liquid cooling, and building MEP systems.
Conduct failure mode analysis (FMEA), root cause investigations, and reliability modeling to meet extreme uptime and efficiency targets.
Collaborate with cross-functional teams on new facility designs, cluster expansions, and continuous improvement of mechanical standards.
Interface with vendors and contractors for equipment selection, fabrication, installation, and commissioning.
Drive design decisions that balance performance, manufacturability, cost, and scalability.
BASIC QUALIFICATIONS:
Bachelor's degree in mechanical engineering or other engineering discipline
PREFERRED SKILLS AND EXPERIENCE:
Previous experience in designing and maintaining industrial HVAC, plumbing, mechanical systems
Proven track record in clean-room design and maintenance, underscoring an understanding of stringent environmental control requirements
Proficiency in implementing maintenance solutions with minimal operational impact
Demonstrated ability to assist technicians with troubleshooting and repair efforts effectively

Capacity to thrive in high-concentration, high-stress environments, with flexibility to work extended hours as required
Attentive to detail and proactive in executing preventive maintenance initiatives
Familiarity with AutoCAD/Revit software applications at a beginner to intermediate skill level
Knowledgeable about relevant industry codes and standards, such as ASHRAE, NFPA, and International Building Code
Previous experience in system design, sizing, specification, and installation, particularly focusing on HVAC chillers, pumps, boilers, air handlers, and HEPA filtration systems
In-depth understanding of HVAC operations, especially in managing chilled water plants operated through air-cooled chillers
Experience with advanced data/monitoring systems like Schneider Building Management Systems is preferred
Strong understanding of construction project management principles, including scheduling, budgeting, and contractor management
ADDITIONAL REQUIREMENTS:
Physical ability to perform hands-on work in various environments, including heat, cold, rain, and tight quarters
Ability to work extended hours and weekends as needed
Ability to travel occasionally as required by project needs
Valid driver's license
Ability to lift up to 25 lbs. unassisted